

Central Dogma of Molecular Biology

Q & A tutorial

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1. What is the central dogma of molecular biology?
2. What are the Characteristics of replication?
3. What are the components required for DNA replication?
4. What are Okazaki fragments and how are they formed?

1. What is the central dogma of molecular biology?

The central dogma of molecular biology describes the fundamental flow of genetic information:

1- Replication

2- Transcription

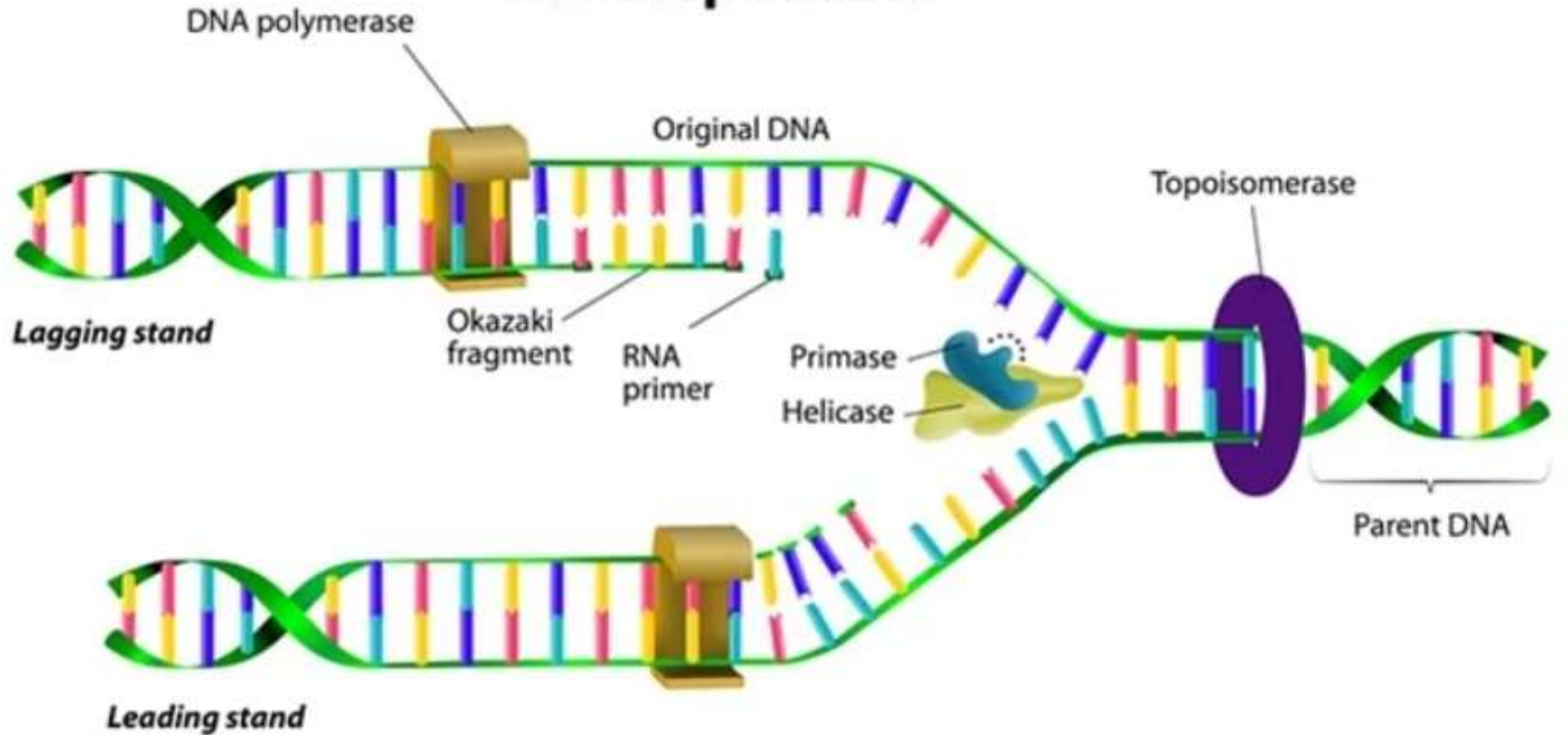
3- Translation

by which the information in genes flows in a UNILATERAL direction: DNA → RNA → protein.

Characteristics of replication

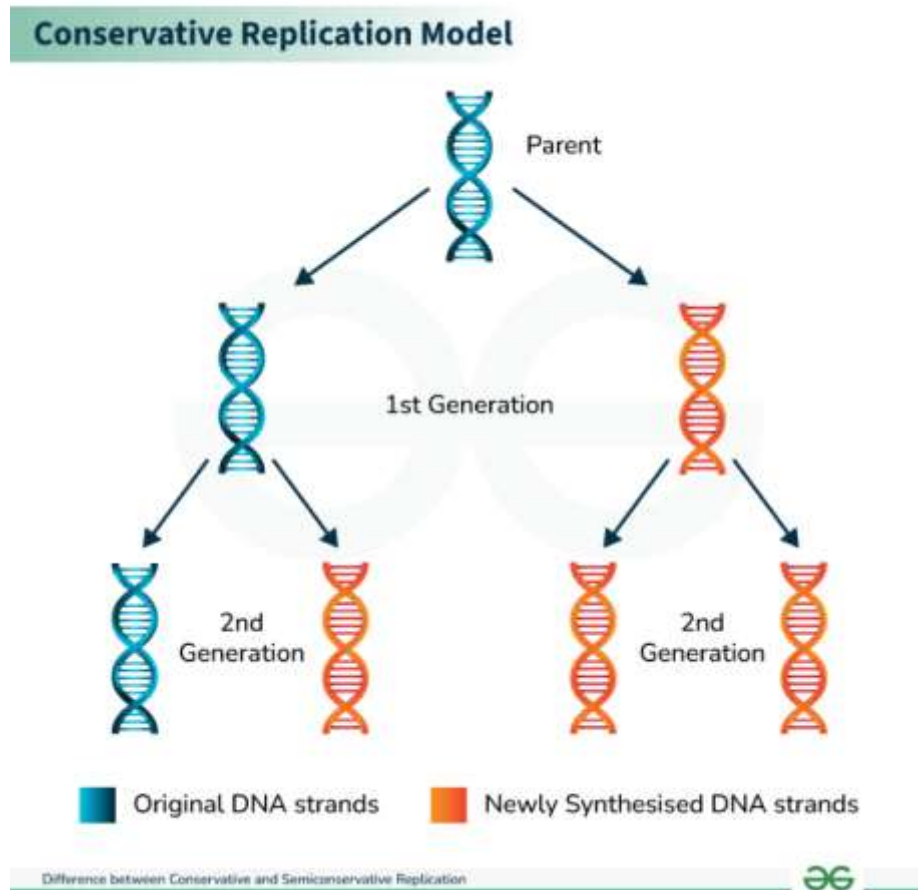
- 1. Starts at replication fork (single in prokaryotes; multiple in eukaryotes).
- 2. Semiconservative.
- 3. Bidirectional.
- 4. Semi-discontinuous (leading strand ; lagging strand).
- 5. Both DNA strands act as template simultaneously.
- 6. Enzymes: helicase, DNA –polymerases, ligase.
- 7. DNA polymerase requires a short RNA primer.
- 8. Synthesis proceed in 5' - 3' direction (reading in 3' - 5').
- 9. Proofreading 3' - 5'.
- Characteristics of replication

DNA replication



DNA replication - Image Credit: Designua / Shutterstock

What is semiconservative replication?



2. What are the components required for DNA replication?

1. **Original DNA template:** DNA replication requires an existing double stranded DNA to copy.
2. **Free DNA nucleotides:** free bases (ATCG) that can be added to the new DNA strand.
3. **Primers:** short strands of nucleotides required by polymerase to build the new DNA strand. DNA polymerase cannot start to extend the DNA strand without a primer to start from.
4. **Enzymes:**
 1. Helicase; facilitates the unwinding of the parent DNA strand.
 2. DNA polymerase; the enzyme required that adds new nucleotides to the new DNA strand.
 3. Topoisomerase; enzyme that prevents supercoiling as the original DNA strand is unwound
 4. Ligase; joins new sections of the DNA strand.

4. What are Okazaki fragments and how are they formed?

Short, discontinuous DNA segments synthesized on the lagging strand during DNA replication, allowing both new strands to be built in the essential 5' to 3' direction as the replication fork opens

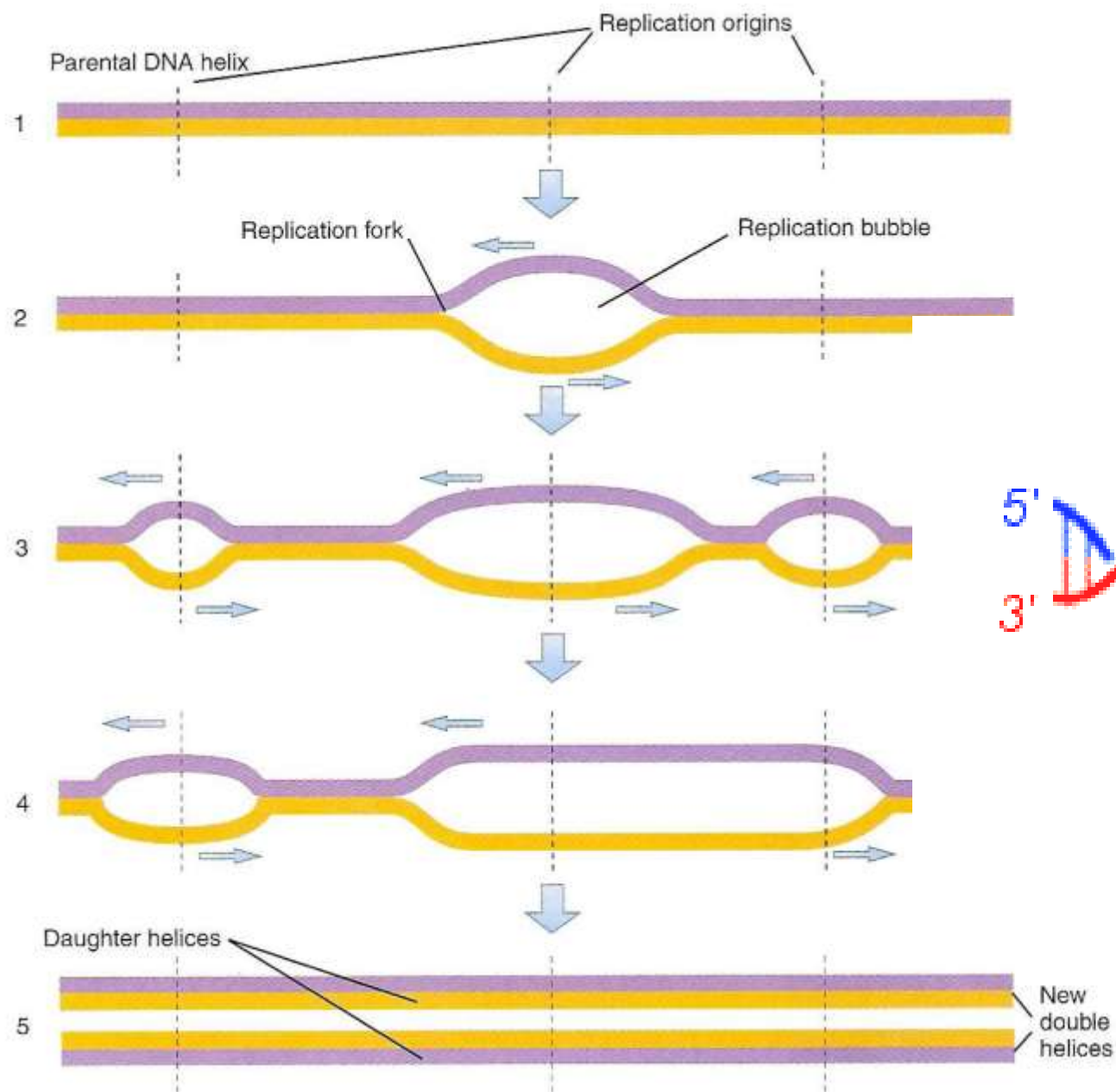
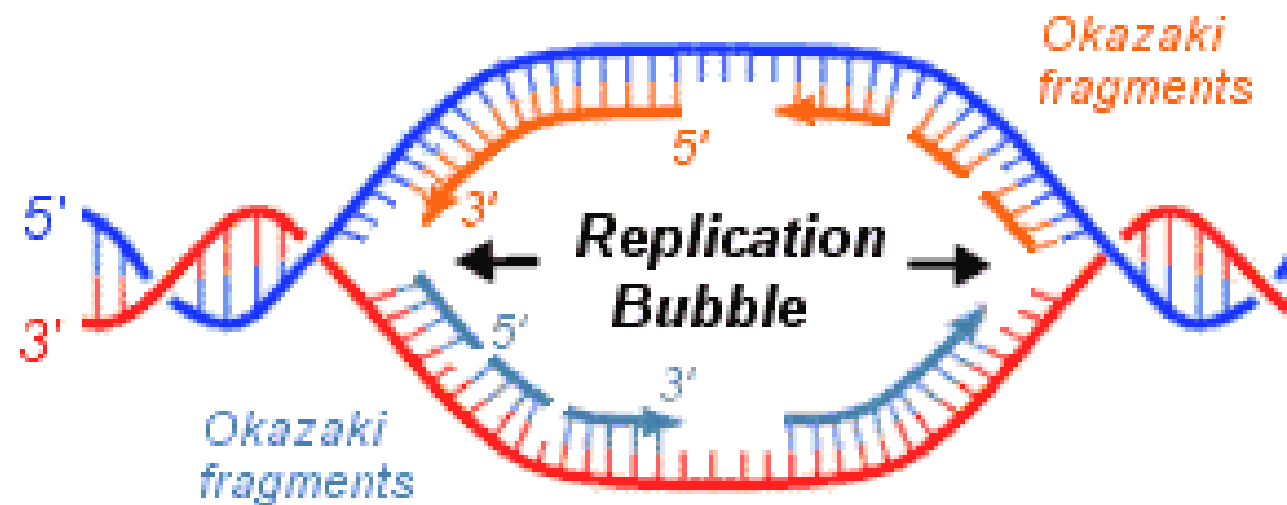


FIGURE 2-6
 Replication bubbles form at multiple points along the DNA strand, allowing replication to proceed more rapidly.



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Protein synthesis - Transcription

1. What are the steps of protein synthesis?
2. What are the key differences between DNA replication and transcription?
3. What are the stages of transcription?
4. What are the 3 main types of RNA?

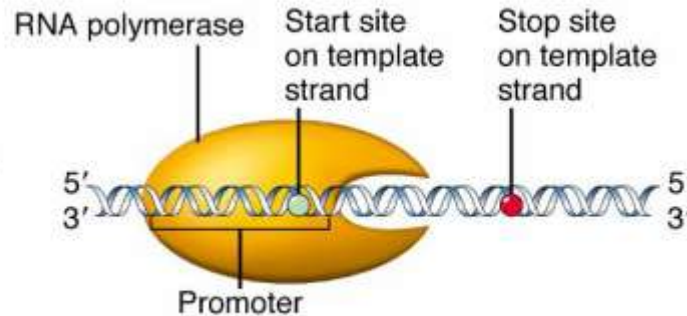
2. What are the key differences between DNA replication and transcription?

- Process of copying DNA to DNA (replication) vs DNA to RNA (transcription).
- Differs from DNA synthesis in that only one strand of DNA, the *template strand*, is used to make mRNA
- Transcription does not need a primer to start.
- Transcription can involve multiple RNA polymerases.

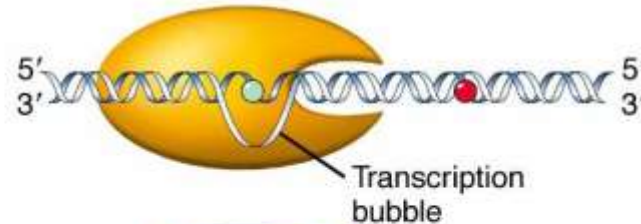
3. What are the stages of transcription?

INITIATION

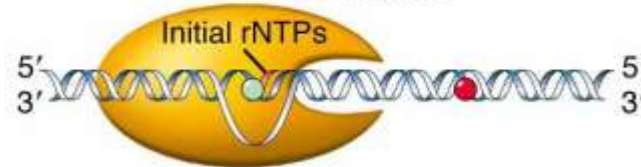
- 1** Polymerase binds to promoter sequence in duplex DNA. "Closed complex"



- 2** Polymerase melts duplex DNA near transcription start site, forming a transcription bubble. "Open complex"

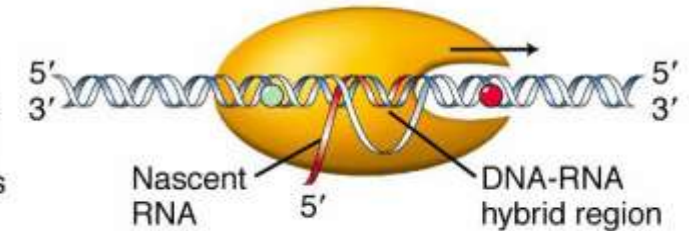


- 3** Polymerase catalyzes phosphodiester linkage of two initial rNTPs.



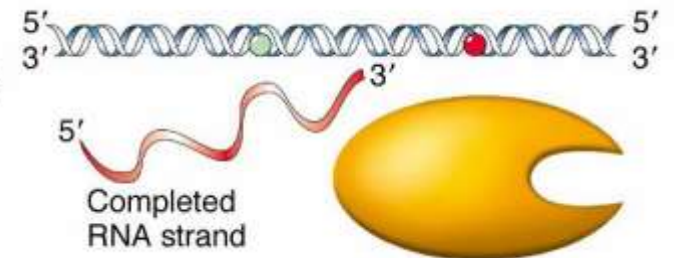
ELONGATION

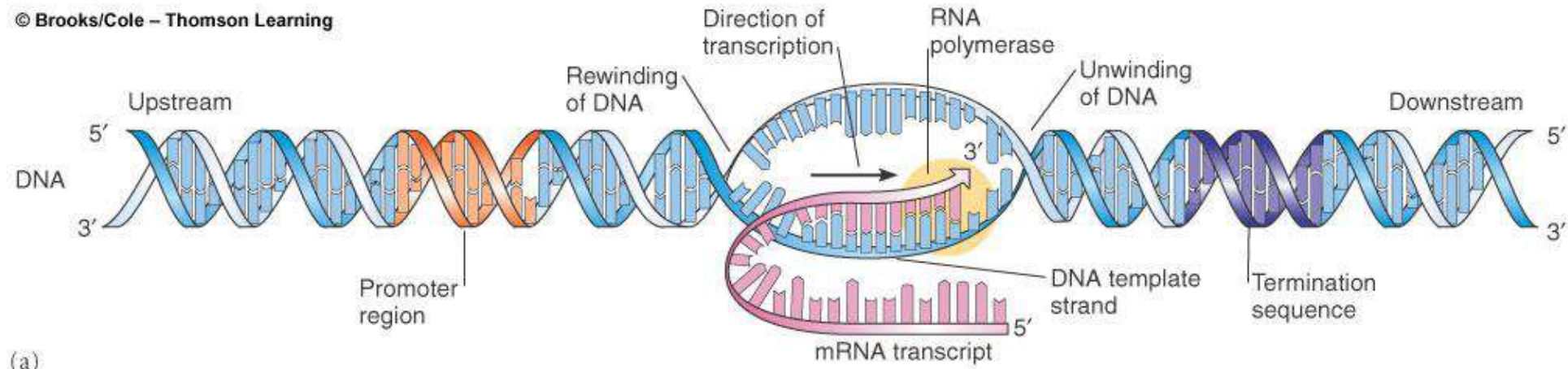
- 4** Polymerase advances $3' \rightarrow 5'$ down template strand, melting duplex DNA and adding rNTPs to growing RNA.



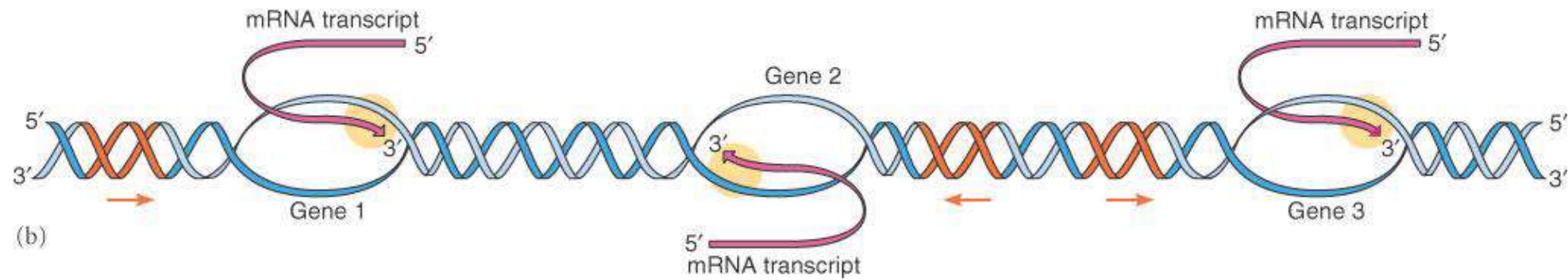
TERMINATION

- 5** At transcription stop site, polymerase releases completed RNA and dissociates from DNA.





(a)



(b)

4. What are the 3 main types of RNA?

- mRNA,
- tRNA,
- rRNA.

Post-transcription Modifications

- Difference between eukaryotes and prokaryotes?
- What are the types of types of eukaryotic mRNA modifications?

Eukaryotic DNA modification

1) Capping:

- Begins when mRNA is about 20-30 nucleotides long.-
- (7-methyl guanosine cap is added to 5' end of mRNA).

Importance:

- 1) Non-capped mRNA can not be recognized by ribosomes.
- 2) Protection against degradation.
- 3) Increase the stability of mRNA.

2) Polyadenylated tail (Poly-A tail)

addition of 100 – 250 adenine nucleotides to the 3' end of completed mRNA.

Importance:

- 1) Help in passage of mRNA from nucleus into the cytoplasm.
- 2) Stabilizes the mRNA against degradation (increase life span of mRNA in cytoplasm)

3) Splicing:

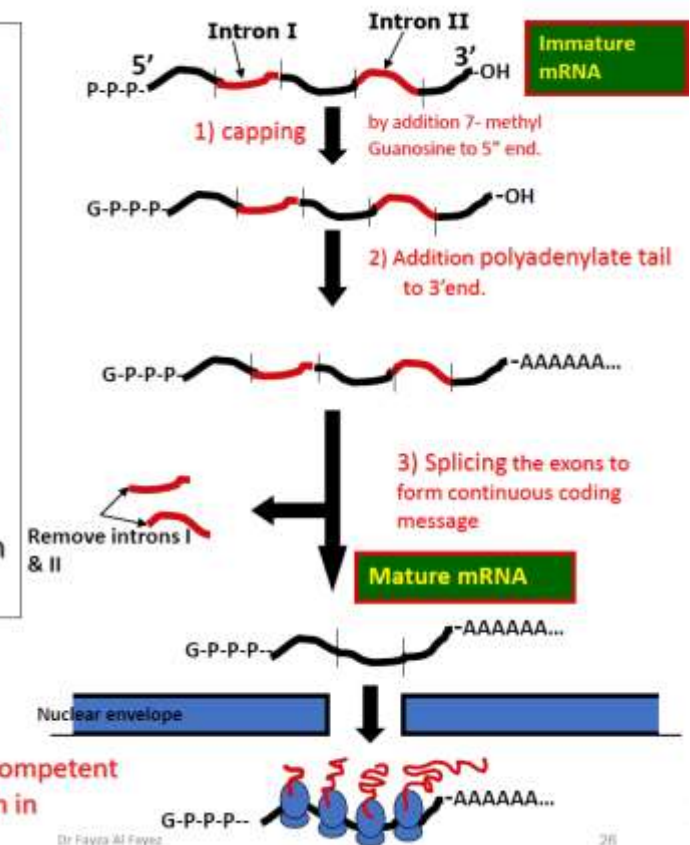
= Means removing the introns (non-coding regions) and splicing the exons (coding regions) together.

- Removal of introns is performed by small nuclear ribonucleoprotein complex enzyme (snRNPC).

Importance:
formation a continuous protein coding message.

mRNA half lifetime:

- about 10 hs. In Eukaryotic cell.
- about 2 mins. In prokaryotic cell.



translation

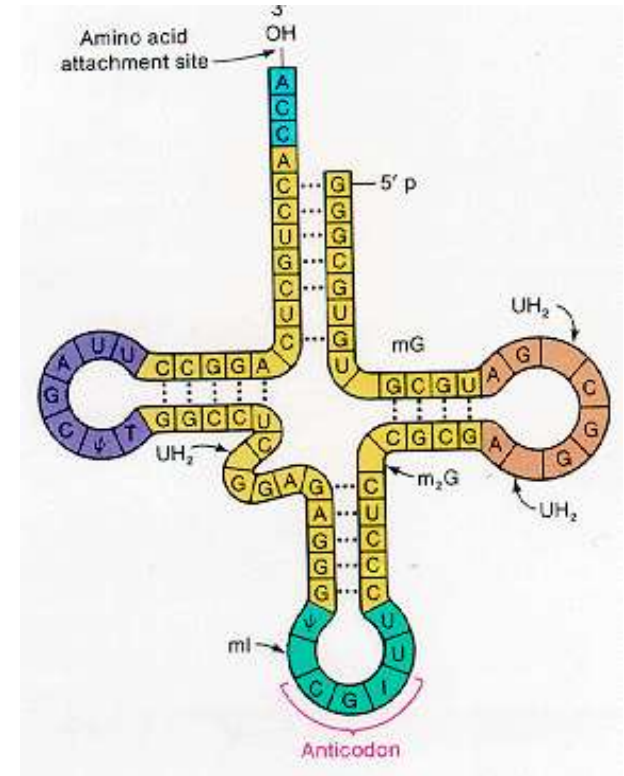
1. What are the components of the translational machinery?
2. What are the types of tRNA?
3. What are the steps of translation?

1. What are the components of the translational machinery?

- mRNA
- ribosomes,
- tRNAs
- and rRNAs.

2. What are the types of tRNA?

- 1) initiator tRNA (carries methionine recognizes & links the start codon; AUG).
 - 2) regular tRNA for other amino acids.
- *Properties:*
- Recognized by amino-acyl-tRNA synthetase & ribosomes.
 - Have amino acid acceptor 3' end.
 - Have anticodon that recognizes and links the complementary mRNA –codon.



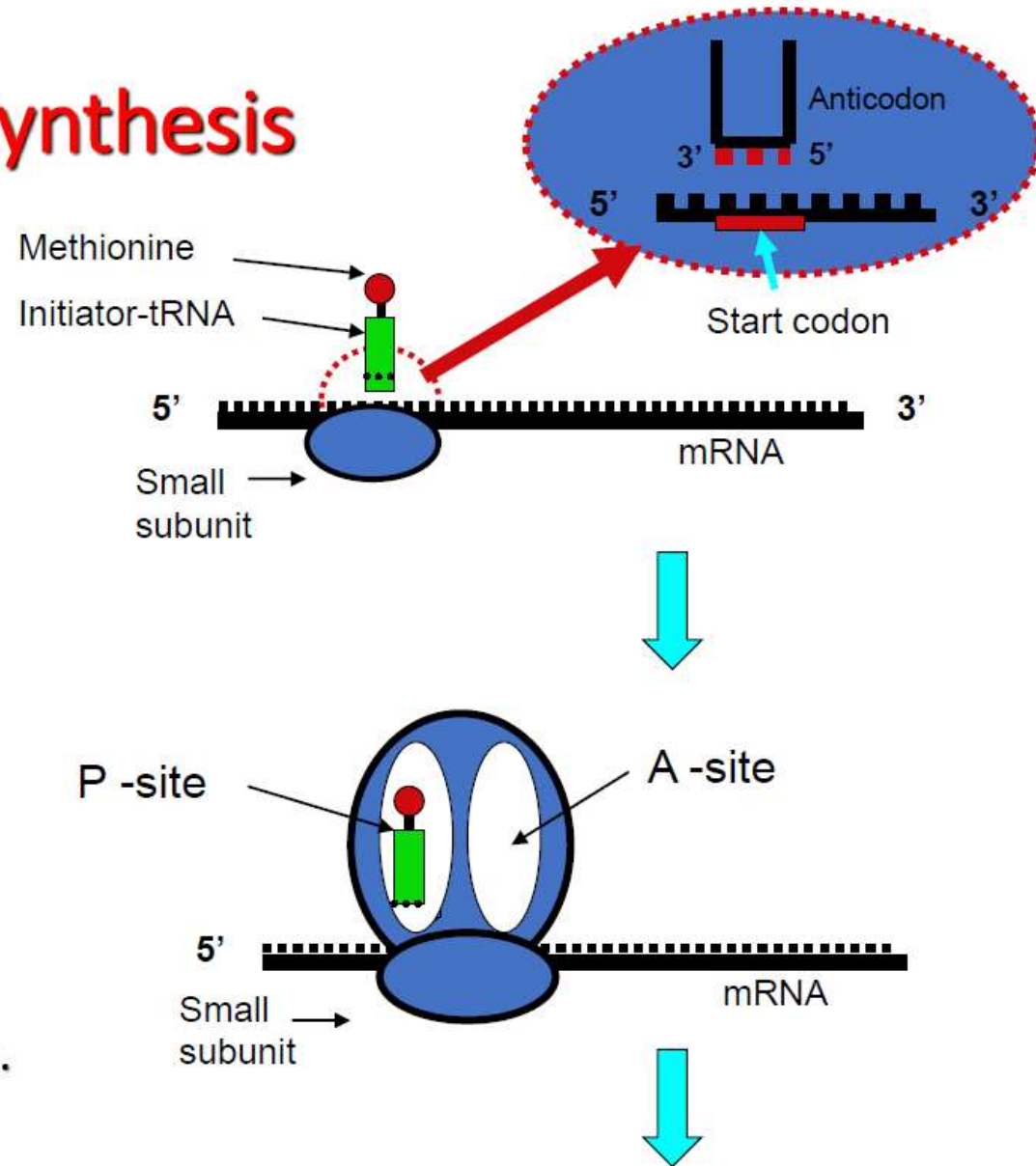
3. What are the steps of translation?

- Initiation
- •Elongation
- •Termination
- •Recognition of STOP codons
- •Usage of release factors

• Steps of Protein Synthesis

1) Initiation

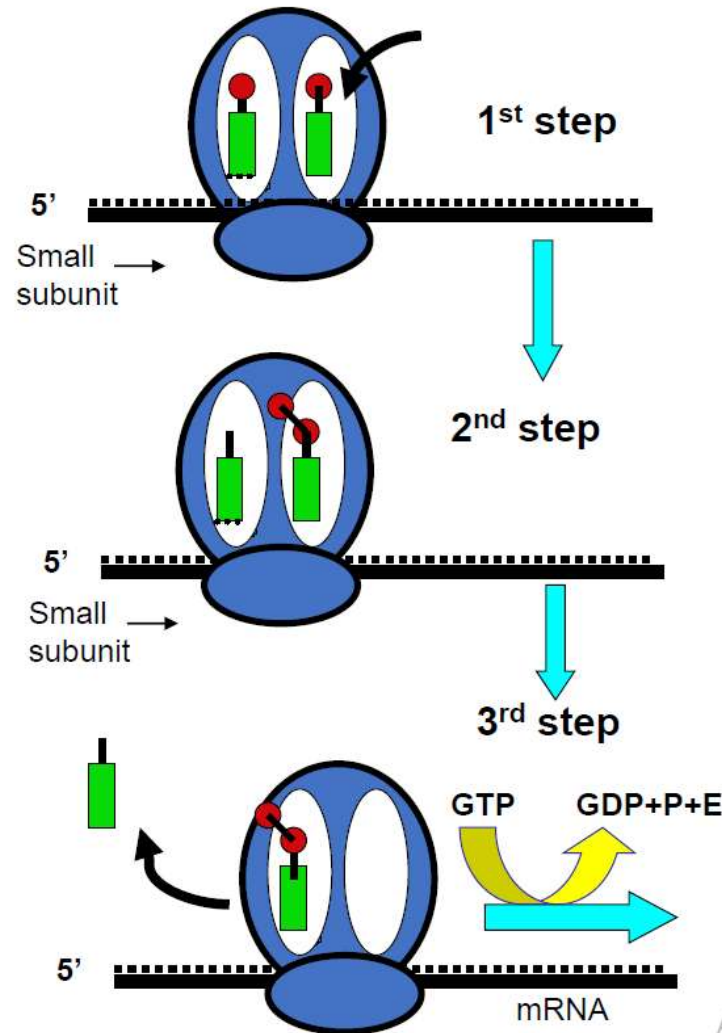
- Begins *when initiator complex* (initiator tRNA & small ribosomal subunit) binds the start codon of mRNA.
- That followed by formation of *functioning ribosome* by binding of large ribosomal subunit to initiator complex.



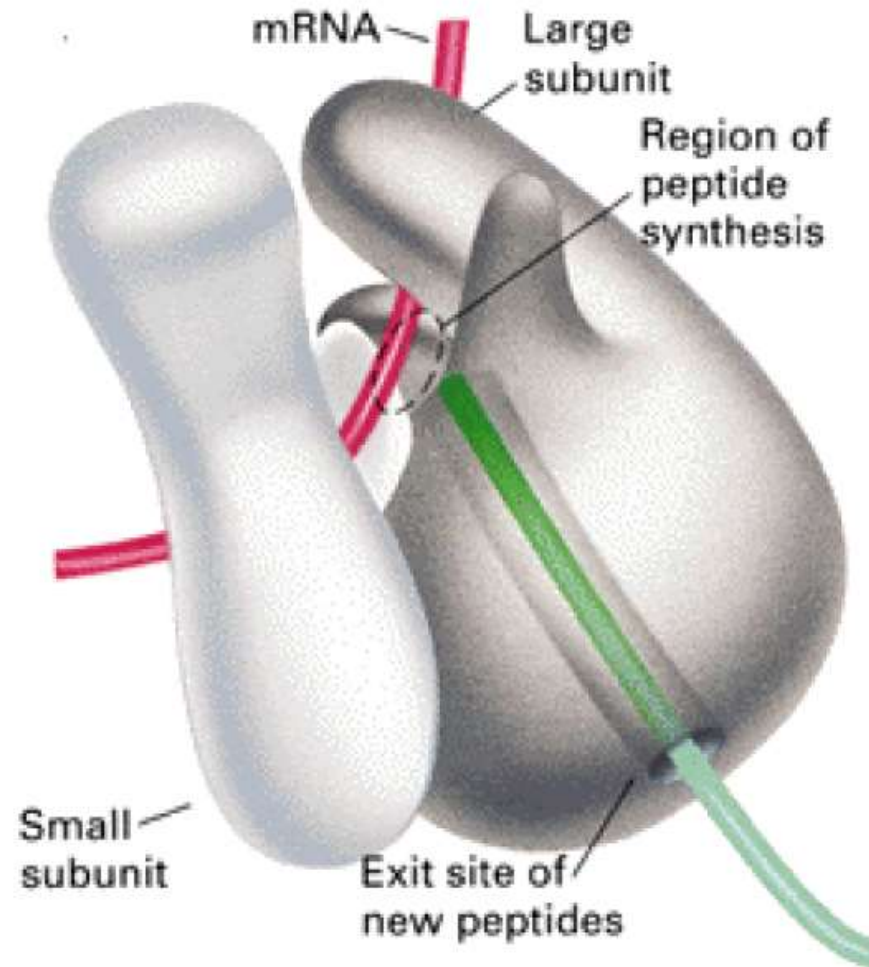
2) Elongation

- 1) *Begins by occupying A- site by 2nd tRNA with its amino acid, that carries anticodon complementary to the next mRNA –codon.*
- 2) *Peptide bond formation; Amino acid is detached from tRNA in P-site & joined by peptide bond to amino acid linked to tRNA in A-site.*
- 3) *Translocation,*
Ribosome moves one codon in direction 5' – 3', that leads to:
 - *release tRNA from P-site to the cytoplasm.*
 - *transfer tRNA & the peptide chain transfer from A- site to P-site.*
 - *A- site become open to receive another suitable tRNA.*

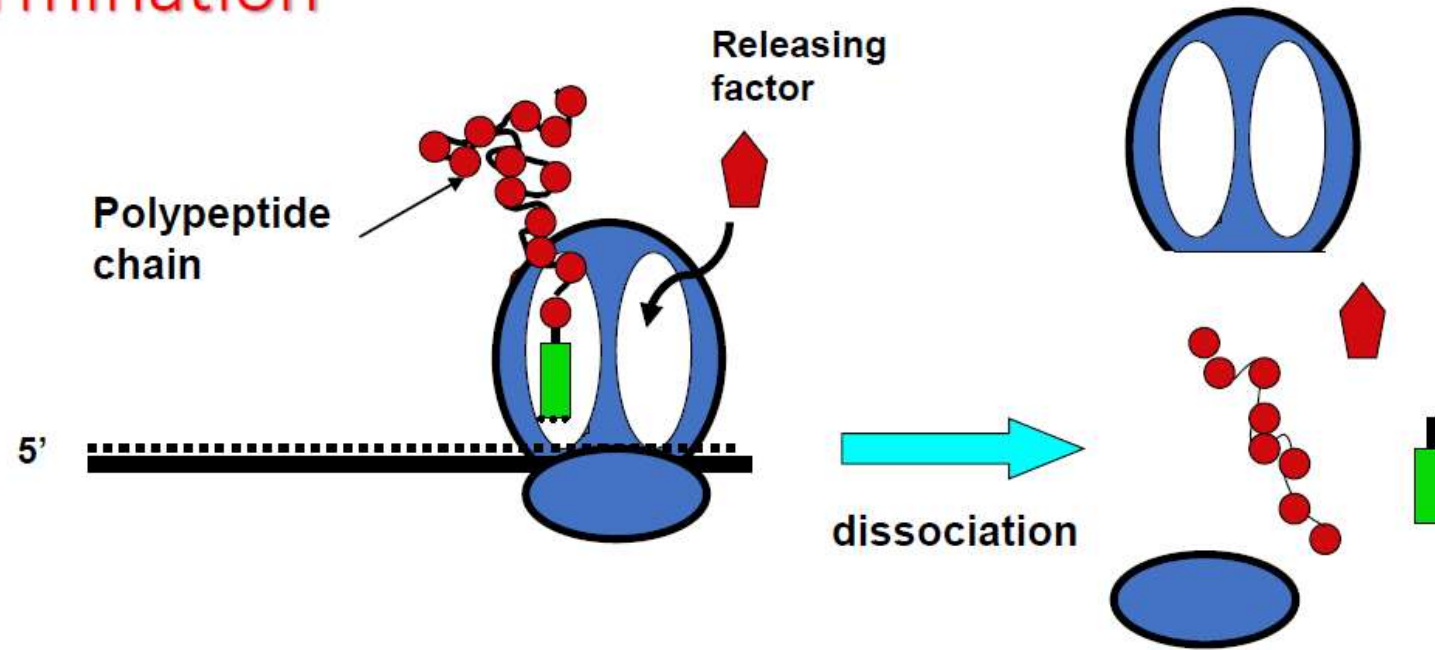
by repeating steps 1, 2, 3 (elongation cycle), the peptide chain elongates till the ribosome reaches the stop codon.



Growing polypeptide chain



3) Termination

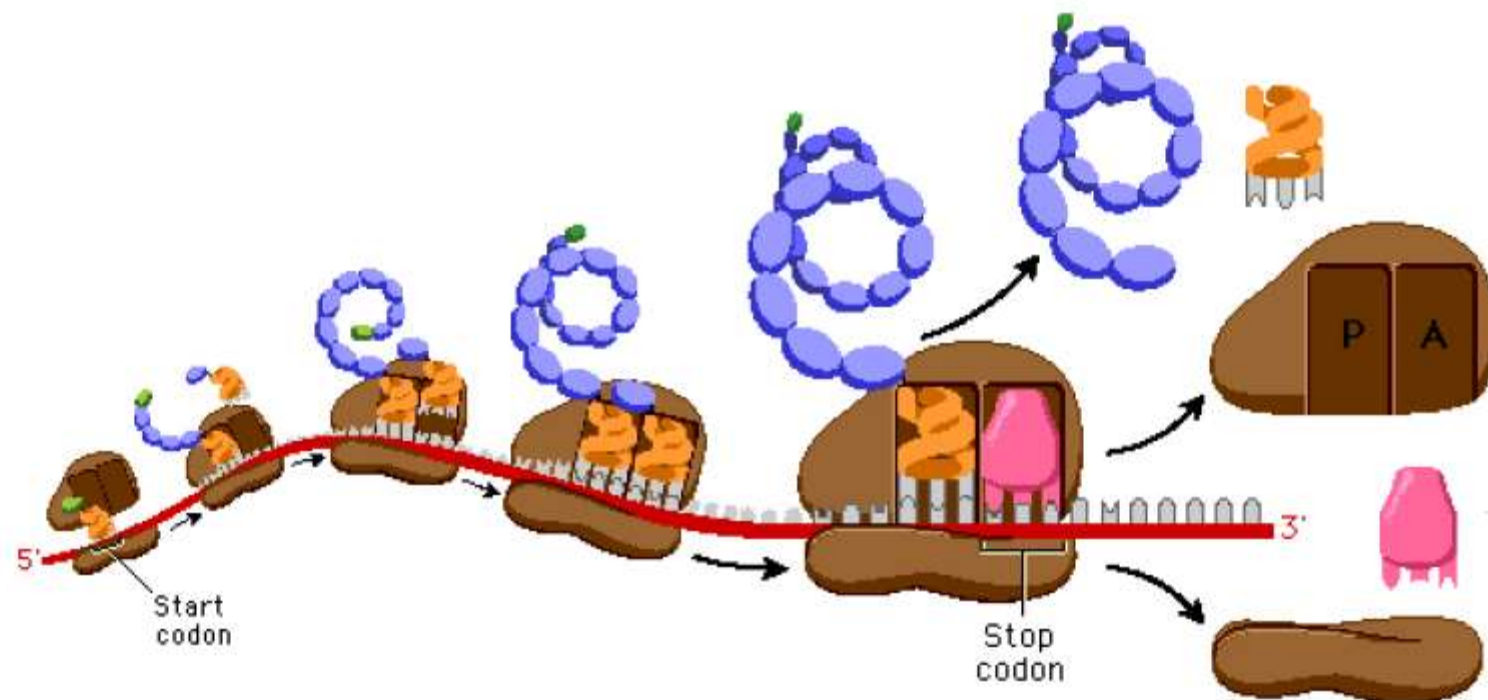


At stop codon:

Release factor recognizes and binds the mRNA - stop codon. It terminates the protein synthesis by releasing:

- Large & small ribosomal subunits
- Polypeptide chain
- tRNA molecule
- Releasing factor

Polysomes



Characters of genetic code

1. **Triplet code** (three bases)
2. **Universal** for all organisms
ex. : UUU = phenylalanine in all organisms.
3. **Redundant, redundancy of genetic code**: more than one code specify one amino acid. (61 codon for 20 amino acid).
ex. : Leucine has 6 different codons.
4. **Reads as a series** of nucleotide.
 - No comma between the codon.
 - The start codon determines the **reading frame**.
- 5) The codon could undergo **mutation**.

Regulation of gene expression

- What is constitutive vs regulated gene expression?
- When does gene expression regulation mostly occur?
- What are some examples of regulation of gene expression?

1. What is constitutive vs regulated gene expression?

- **Constitutive** gene expression refers to gene expression that is constantly present and active. They are also known as house-keeping genes. These genes are constantly expressed under normal growth conditions.
- while **regulated** gene expression refers to gene expression that is controlled in response to the needs of a cell or organism. The activity of gene expression can vary according to cell/tissue type or in different stages of development such as Hemoglobin F is active in fetal stage of life then inhibited after birth when hemoglobin A is activated.

2. When does gene expression regulation mostly occur?

- Gene expression in prokaryotes is regulated only at the transcriptional level, whereas in eukaryotic cells, gene expression is regulated at the epigenetic, transcriptional, post-transcriptional, translational, and post-translational levels.

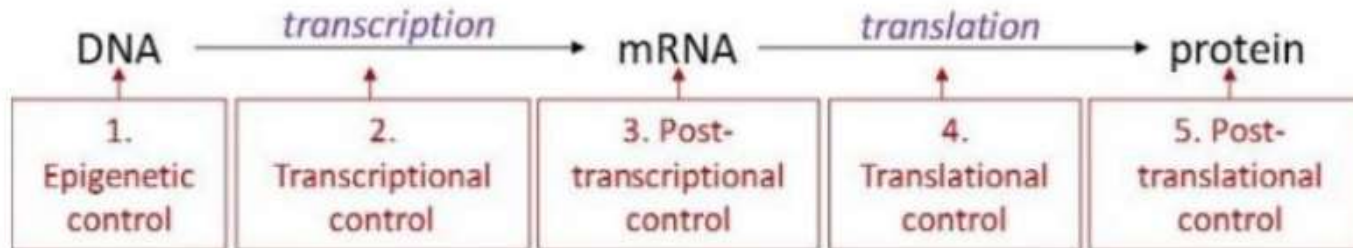


Figure 17.6 Regulation of gene expression in eukaryotes can occur at five different levels. Here, the Central Dogma is diagrammed with arrows showing where each type of eukaryotic regulation of gene expression interrupts it.

3. What are some examples of regulation of gene expression?

- Modification of Histone Proteins (epigenetic)
- The Promoter and Transcription Factors (transcriptional)
- Enhancers and Repressors (transcriptional)
- Alternative RNA splicing (Post-transcriptional)

Questions?

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